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Mercury Sample Return Mission Design Utilizing Innovative Systems and Technologies

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Abstract

Mercury is one of four terrestrial planets, and the smallest planet in the solar system, known to have a significant metallic core and a high metal to silicate ratio. However, the reason for the planet's specific composition and formation is yet to be explained. While many theories attempt to justify these facts, none can be confirmed without further investigation, such as one provided by a sample return mission. The science collected from this mission will allow for validation of theories about the formation of Mercury and possibly the Solar System. By investigating theories about Mercury's formation, more information will be uncovered about the Earth's Moon as there are quite a few similarities between the two bodies. These similarities will help establish a newer, more grounded theory about the origin of the Moon and its effects on Earth. Additionally, learning about Mercury could provide information on how the Earth might evolve over time. Sending a spacecraft to Mercury has many challenges, orbital mechanics being one of the biggest. As Mercury is the closest planet to the Sun, its orbit is the most affected by the Sun's gravity, making it difficult to manoeuvre a spacecraft into a stable orbit. Moreover, going to Mercury is a complicated and long-duration mission requiring gravity assists and complex trajectories. Thermal control is another challenge since the spacecraft must be protected from Mercury's extreme temperatures. In this context, this research attempts to deal with these difficulties through innovative analyses of technologies to develop a feasible plan to perform such a mission. The study presents trades of different technologies and approaches for subsystems. Due to the previously discussed challenges imposed by Mercury, employing an innovative approach to the technologies will allow for designing a mission optimal for combatting challenges. The paper also discusses the mission design tools and Model-Based Systems Engineering (MBSE) approaches used to validate results. Please note that this work is submitted under the guidance of the Space Exploration Project Group (SEPG) of the Space Generation Advisory Council (SGAC) as part of the ACHIEVED Initiative, which stands for Assembly for Concepts in Human Interplanetary Exploration with Various Extraterrestrial Designations. This initiative aims to enable students and young professionals from diverse technical and cultural backgrounds to work and collaborate on a space mission design project.

Keywords: Mercury, Sample, Mission, Model Based Systems Engineering, Innovation

Acronyms/Abbreviations:

SGAC - Space Generation Advisory Council
RAISE - Research Assembly for Innovative Projects in Space Exploration
ConOps - Concept of Operations
MESSENGER - Mercury Surface, Space Environments, Geochemistry, and Ranging

NASA - National Aeronautics and Space Administration
JAXA - Japanese Aerospace Exploration Agency
MMX - Martian Moons eXploration
GNC - Guidance Navigation and Control
IMU - Inertial Measurement Unit
3-DOF - Three Degrees of Freedom
TDS - Terminal Descent Sensor

GPHS-RTG - General Purpose Heat Source
Radioisotope Thermoelectric Generator
DSS - Digital Sun Sensor
FOV - Field of View
SRU - Shunt Regulator Unit
PDB - Propulsion Diode Box
SEP - Solar Energy Particles
DoE - Department of Energy
ASRG - Advanced Stirling Radioisotope Generator
BOM - Beginning-of-Mission
ASC - Advanced Stirling Converter
GRC - Glenn Research Centre
CDH - Command and Data Handling
EPS - Electrical Power System
ADCS - Attitude Determination and Control System
SMRTG - Segmented-Modular RTG
O-HAS - Optical Hazard-Avoidance Sensor
M-HAS - Multifunction Hazard-Avoidance Sensor
CAD - Computer Aided Design
MOSIF- MMO Sun shield and InterFace
MPO- Mercury Planetary Orbiter
HT-MLI - High-Temperature Multi-Layer Insulation
RTC - Real-Time Commands
SSC - Stored Sequence Commands
RTU - Remote Terminal Unit
CDHMu - Command and Data Handling Management unit
SSMM - Solid-State Mass Memory
PM - Processor Modules
SDST - Small Deep-Space Transponder

Nomenclature:

Δv - Delta Velocity
 v_e - Exhaust Velocity
 m_0 - Wet Mass; Initial Mass Pre-Burn
 m_f - Dry Mass; Mass Post-Burn
 I_{sp} - Specific Impulse
 g_0 - Gravitational Constant

1. Introduction

Mercury remains to be the least explored planet in our solar system. Nevertheless, the scientific community knows that collecting and analysing a Mercury surface sample can lead to multiple new scientific discoveries. Learning about the surface composition can validate theories on Earth's Moon's formation and how the Earth may evolve.

On October 20th, 2018, the European Space Agency (ESA) and the Japanese Space Agency (JAXA) successfully launched the BepiColombo Spacecraft on a seven-year journey to Mercury, followed by one year of operations after reaching the planet. This mission will provide many answers to the questions of the scientific community through the data collected by scientific

instruments. However, the spacecraft will be orbiting Mercury collecting scientific data and will not be collecting and returning a sample.

If humanity wants to continue exploring our solar system, the next generation of space missions will require innovative designs and new technologies that push the limits of what has been done, enabling us to advance our experience as space explorers. Therefore, the Space Generation Advisory Council's (SGAC) Research Assembly for Innovative projects in Space Exploration (RAISE) team has designed a mission that explores collecting a Mercury surface sample and transporting it back to the Deep Space Gateway. Additionally, this mission will build upon the science performed by the BepiColombo mission, which would have collected more details on Mercury's surface and composition.

The mission design process prioritised innovation and mission success as is evident in the trade-offs and decisions driving the higher-level design. Innovative strategies across subsystems included using newer technologies, maximising modularity, and utilising new tools for design.

By researching the latest technologies emerging in the space industry, the team incorporated new technological concepts that would be available when the mission launches, around the 2040s. Newer technologies address common limitations faced in deep space missions, such as power production and mass limitation. On the other hand, modularity results in a flexible mission that can adjust the spacecraft to optimally perform in different mission phases. Lastly, digital tools support design activities by improving the quality of systems engineering processes. For example, Model Based Systems Engineering (MBSE) allowed the team to visualise the different subsystems' architectures and define interfaces and interactions all in a single diagram. The subsequent sections will give more details on the different design strategies.

2. Mission Objectives

The mission has two high-level requirements which can be met by collecting a surface-level sample. Surface level samples require less complexity and mass compared to deeper core samples.

Table 1. Mission Requirements

ID	Name	Description
L1.1	Mercury Formation	The Mission shall result in the increased knowledge of Mercury's formation through sample collection.

L1.2	Mercury Magnetization Formation	The Mission shall result in the increased knowledge of the formation of Mercury's magnetised atmosphere.
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2.1 Requirement L1.1

The first requirement is satisfied through the collected samples. Depending on the nature of their composition, the scientific community could make conclusions on how Mercury was formed. Currently, two potential theories explain the planet's formation: (i) fractionation of metal from silicate during formation and (ii) fractionation after collisional processes. The first theory involves processes before accretion that remove silicate or accumulated metal in the inner part of the solar nebula. Conversely, the second theory involves collisional processes where one or more collisions between a proto-Mercury and other bodies ejected a significant portion of material from the silicate portion of proto-Mercury. Therefore, learning about the composition of the surface of Mercury could help validate one of these theories. [1]

2.2 Requirement L1.2

The second requirement refines the previous one by focusing on the metallic constituents found in the sample since determining the nature of the metallic constituents can help determine the nature of Mercury's magnetosphere. The landing site was chosen with this in mind. Additionally, a magnetometer is included in the Orbiter Segment to help characterise the magnetosphere. [1].

3. Mission Concept of Operations (ConOps)

The Mission ConOps includes all activities required to perform the mission. In our MBSE approach, the Mission ConOps is defined by the Operational Architecture, which displays the operational entities, activities, and interactions when performing the mission, resulting in 7 different processes that lead to mission completion. The diagrams used to display the ConOps can be found in Appendix A and can be used to reference the different mission phases.

3.1 ConOps Overview

The mission is divided into multiple phases and is expected to have a long duration. It starts in the 2040 decade utilising a launch vehicle capable of providing a boost for Earth escape to provide the delta-Vs required. Earth escape requires a spacecraft to perform a delta-V manoeuvre to escape the gravitational pull from the Earth and by requiring the launch vehicle to perform this manoeuvre the spacecraft is alleviated of the propellant requirement. The desired launch vehicle, based on the need for this capability and the mass of the

spacecraft, is a Falcon Heavy class. However, we expect launch vehicle capabilities to improve in the future so that other options can be considered.

From Earth's escape, the spacecraft then reaches Venus, where it will get gravity assists to reach Mercury with a lower delta-V expenditure. Gravity assists can decrease or increase a spacecraft's velocity to give it an orbital mechanics boost. After reaching Mercury, the spacecraft will be in a parking circular orbit of an altitude of 200 km. For forward work, the team would analyse this orbit to better determine its viability.

The Lander and Ascent segments will then descend as an integrated unit to a landing location near the south pole for landing operations. After collecting a sample, the Lander will remain on the surface while the Ascent Segment ascends to rendezvous with the Orbiter Segment in its parking orbit and dock.

The return trajectory will also involve a Venus gravity assist, but it will aim to reach the Deep Space Gateway for sample return.

For each manoeuvre, the team did higher-level calculations and utilised delta-Vs from previous studies, such as the JHU decadal survey [2].

3.2 Landing Location

The chosen landing location is at around 85° S and 200° E (see Fig. 2), in an area outside of the large crater in the south pole. This area provides desirable thermal conditions and its proximity to a crater that is theorised to contain ice in its permanently shadowed regions. This site has been deemed an area of interest from observations collected by the Mercury Surface, Space Environments, Geochemistry, and Ranging (MESSENGER) spacecraft, notably the high radar reflectivity that could indicate ice deposits [2]. The craters in this area are also thought to be a result of volcanic activity, similar to the Moon, providing a valuable sample to meet the mission requirements.

Additional imagery and data from the BepiColombo mission could provide valuable insight into choosing a different landing location. However, the Orbiter Segment is expected to have cameras to provide supplementary imagery data that could change the landing location.

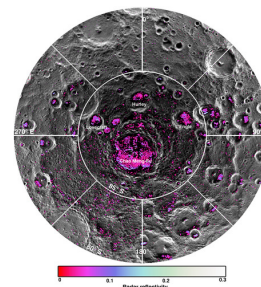


Fig. 1: Mercury South Pole from MESSENGER
Showing Areas of High Radar Reflectivity

4. Mission Segments

The RAISE Mercury Sample Return Mission consists of three space segments: an Orbiter, Lander, and an Ascent Segment (see Fig. 3).

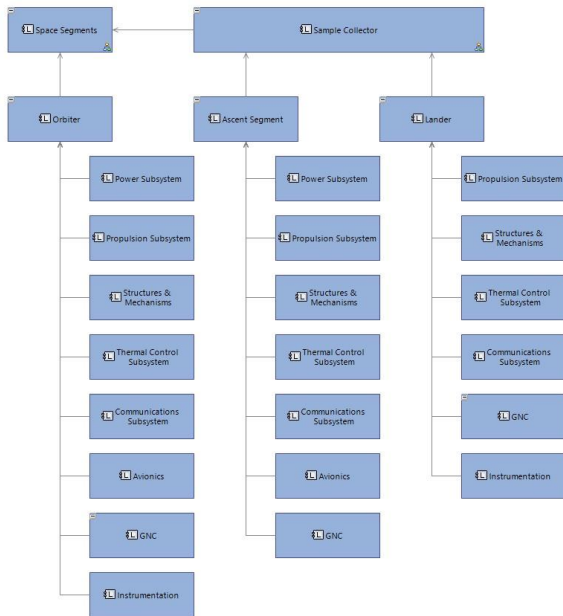


Fig. 2: Space Segment Breakdown

Maintaining a low mass was an essential design driver and the primary constraint since returning a sample from Mercury is a primary mission objective. As a result, the vehicle design focused on minimising the structure and scientific instruments, having the Lander rely on subsystems on the Ascent Segment for some functionality (see Fig. 2). Additionally, the operations at the surface shall last approximately two days, the minimum amount of time needed to collect a sample, driving down the power required and thus reducing the total dry mass.

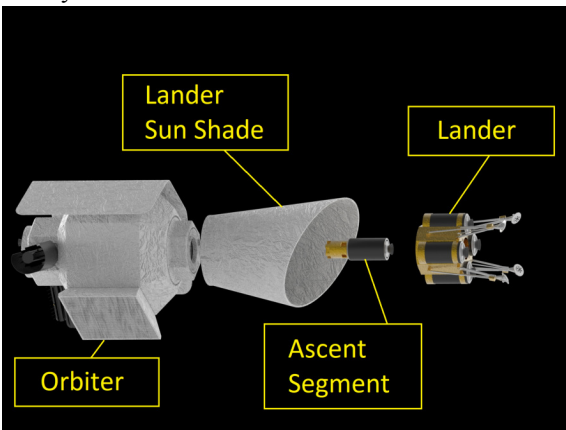


Fig. 3: Spacecraft Computer Aided Design (CAD) Assembly

The mass budget utilises a 30% margin, resulting in a total of 4021 kg at launch. This mass is within the launch vehicle lifting capacities of the Falcon Heavy. The mass budget is in Tables 2 and 3, showing the mass per subsystem and mass per mission phase, respectively.

Table 2. Mass (kg) Summary for Subsystems

	Lander	Ascent Seg	Orbiter
Structure	64	33	231
Payload	35	7	30
Thermal	12	3	75
Power	12	50	191
CDH	-	11	11
Comms.	2	3	23.3
GNC	7	-	20
Propulsion	80	20	71
Dry Mass	212	127	652
Propellant	1400	290	412
Total w/ Propellant (CBE)	1612	417	1064
Marg. 30%	483	125	319
Total (MEV)	2095	542	1384

Table 3. Mass (kg) MEV Summary for Mission Phases

	Lander	Ascent	Orbiter	Total
Phase 1: Cruise	2095	542	1384	4021
Phase 2: Landing Operations	212	542	-	754
Phase 3: Return Cruise	-	127	1126	1253

4.1 Orbiter Overview

The Orbiter Segment serves primarily as a transport vehicle for the Lander and the Ascent Segment during the cruise, orbital, and return phase. During the cruise phase, the Lander and Ascent Segment are integrated into a single unit, with the Ascent Segment located within the Lander.

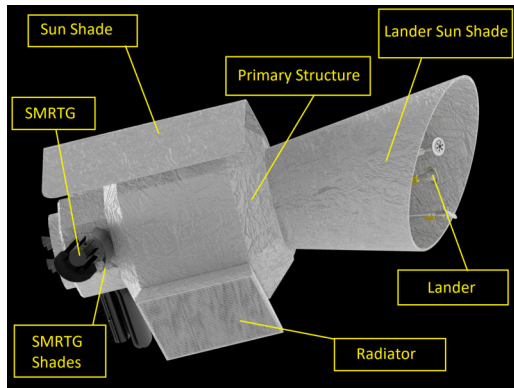


Fig. 4: Orbiter CAD Model

As seen in Fig. 5, the primary function of the Orbiter is the orbital manoeuvres needed to complete the mission. Additionally, during the cruise and return mission phases, the Orbiter communicates with the ground station to monitor the mission. During the return phase, it performs the docking sequence to the Deep Space Gateway orbiting the Moon. Other functions include power production and radiation protection.

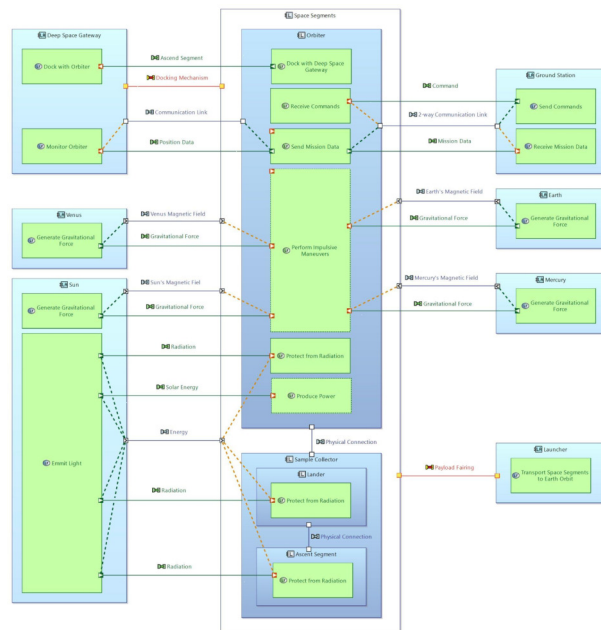


Fig. 5: Orbiter Functional Overview

Fig. 5 also shows the primary interfaces between the Orbiter and external elements including different gravitational fields, the light emitted by the sun, communication links, and physical interfaces that connect the spacecraft to the Launch Vehicle and Deep Space Gateway.

In terms of scientific instruments, the Orbiter has a magnetometer and allocated mass for additional scientific instruments to perform research during the

cruise phase. Its primary structure is hexagonal with two Segmented-Modular RTGs (SMRTG) and two primary sun shades (as seen in Fig. 4). It also has two docking interfaces: a 1.2m-diameter interface for docking with the Deep Space Gateway and Lander, and a smaller docking interface with a cylindrical docking compartment located within the larger interface to connect the Ascent Segment. The Ascent Segment and Lander are connected to make a single unit during the launch and cruise. This Lander and Ascent Segment unit is attached to the orbiter using the two docking interfaces. In this configuration, the Ascent Segment is inside the Lander and locked to the Orbiter through the smaller docking interface to prevent potentially hazardous movement due to forces felt during mission operations. After sample collection, the Ascent Segment again docks with the Orbiter through the small docking interface. At the end of the mission, the large docking interface is used to dock the Orbiter and Ascent Segment to the Deep Space Gateway.

4.2 Lander & Ascent Segments Overview

For more than half of the mission, the Lander and Ascent Segment function as a unified system. Both segments start to operate independently from the Orbiter after undocking at the Mercury parking orbit for landing.

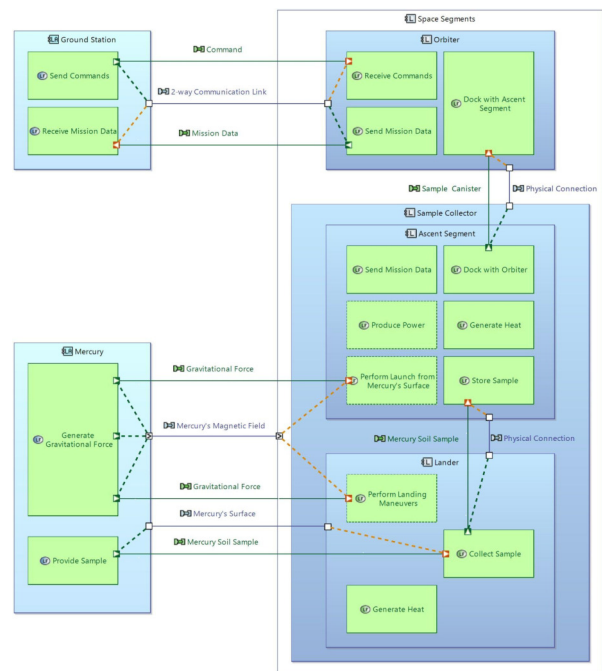


Fig. 6: Lander & Ascent Segment Functional Overview

While the Ascent Segment can function independently, the Lander relies on it for power production and data handling. As seen in Fig 6, their main functions include performing the landing

manoeuvres, collecting the sample, and transporting it back to the Orbiter after docking post-ascent. Other significant functions include power production and heat generation since the Mercury surface will have low temperatures. Lastly, during surface operations, the primary interfaces include the physical interfaces that connect the segments and Mercury's gravitational and thermal environment.

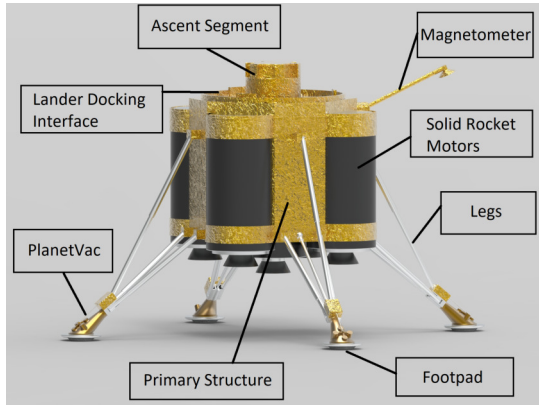


Fig. 6: Lander & Ascent Segment CAD

The Lander is a four-legged, octagonal structure with four solid motors. In addition, the Lander has four PlanetVacs, a magnetometer as the primary scientific instrument, and a central compartment used to house the Ascent Segment. The Ascent Segment returns samples from the surface of Mercury to the Orbiter. Additionally, the data collected from its magnetometer is transferred to the Orbiter when it re-docks post-ascent. Its subsystems, docking mechanism, and sample container are all contained within a basic cylindrical structure. In addition, the Ascent Segment has a detachable solid motor that provides the necessary thrust for ascent from the surface of Mercury.

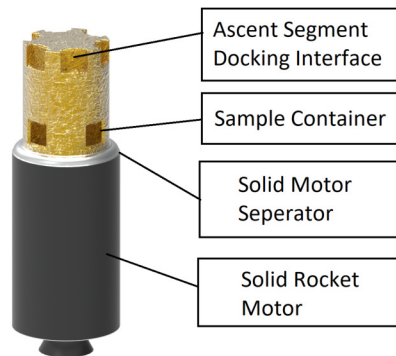


Fig. 7: Ascent Segment CAD model

4.2.1 Sampling Mechanism & Science Equipment

The primary function of the Lander is to collect samples from the landing site and transfer them to the

Ascent Segment for return to the Orbiter. Due to the mass constraints of a Mercury sample return mission, a sampling system for such a mission should be lightweight and capable of collecting a significant volume of samples. However, acquiring a sample from Mercury's surface is not straightforward since collecting deep samples from the subsurface causes a mass increase. Therefore, after a thorough review of existing extraterrestrial regolith-sampling technologies [8][9], the PlanetVac sampler [7] became the sampling system for the Lander, including four PlanetVac collection units as part of the Lander.

The PlanetVac sampler is a robust pneumatic sampling device modified to P-Sampler [5] by HoneyBee robotics and later tested with Masten Space Systems' Xodiac rocket in 2018 [6]. This system was also selected for launch by the National Aeronautics and Space Administration's (NASA) Commercial Lunar Payload Services Program in 2023 and the Japanese Aerospace Exploration Agency's (JAXA) Martian Moons eXploration (MMX) mission in 2024 [7].

As previously mentioned, in addition to the sampling mechanism, the Lander includes a magnetometer to study the magnetosphere of Mercury during its surface operations. Modularity and allocation of the command and data handling subsystem to the Ascent Segment allow the samples and acquired data to be transferred to the Orbiter via the Ascent Segment upon docking.

4.2.1.1 PlanetVac Sampling System Description

PlanetVac has three major components: the Control Box, Tubes, and Sampling Head. [5] The Lander's footpad is integrated with the Sampling Head, similar to the Xodiac rocket test run setup [6]. The Control Box of the P-Sampler version [5] has been modified by removing its sample canister and installing it within the Lander primary structure. A sample transfer tube was added with flexible joints at either end to offset landing forces. The sample transfer tube is attached to the sample canister.

4.2.1.2 Sample Storage Procedures

The PlanetVac sampling head (Fig. 8 Circle 1) will transfer the gathered samples to the sample canister located at the bottom of the primary structure (Fig. 8 Circle 2). After sample completion, the sample canister is transferred into the sample transfer box (Fig 8. Circle 3). The sample transfer box is then moved to the top of the primary structure using motors, reaching the top of the Ascent Segment (Fig. 8 Circle 4). The sample canister is then moved into the Ascent Segment sample container using an actuator (Fig. 8 Circle 5), where the sample canisters will be safely stored for the remainder of the mission.

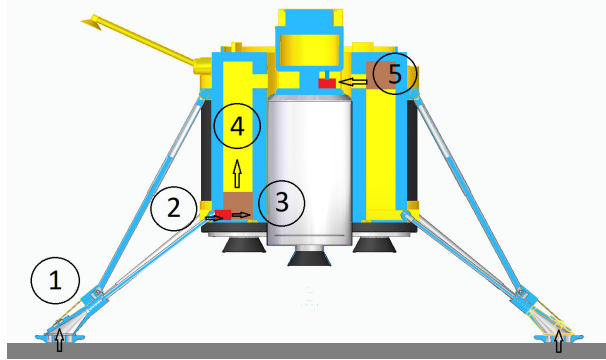


Fig. 8: Lander Sample Collection Procedures

4.2.2 Ascent Segment Procedures

After completing the two-day surface science mission and having the sample canisters transferred and secured, the Ascent Segment will detach from the Lander and ascend using its solid motor to the Orbiter in its Mercury parking orbit. The Ascent Segment then docks to the Orbiter through the smaller docking interface and discards the solid motor.

4.2.2.1 Sample Transfer Mechanism

The Ascent Segment's docking system is within the Orbiter's larger docking interface. This interface has a conical shape to aid the docking of the Ascent Segment. Once the Ascent Segment has been positioned in the cylindrical docking cavity, locks in the sliding mechanisms (Fig. 9) attach the Ascent Segment to the sliders (Fig. 9 Circle 1). Afterwards, the Ascent Segment and sliders slide towards the back of the cylindrical docking cavity (Fig. 9 Circle 2), where Iris locks seal the docking interface (Fig. 9 Circle 3, 4). This process must be executed in reverse order when transferring the samples from the Orbiter to the Deep Space Gateway.

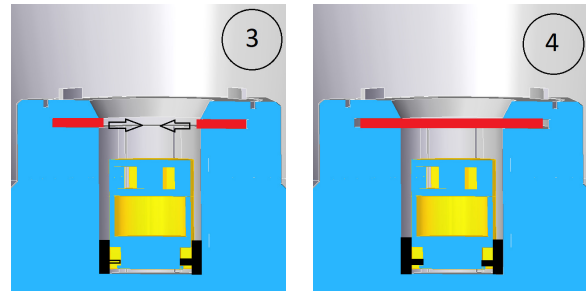
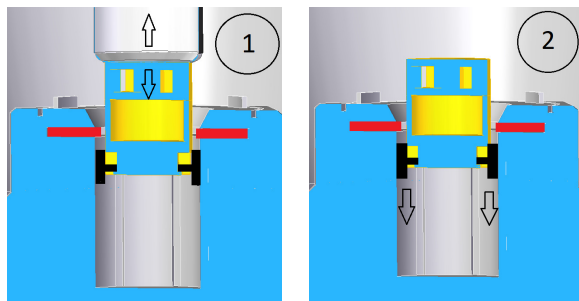


Fig. 9: Orbiter Sample Storage

5. Propulsion

The propulsion system was a key driver of the design of the entire spacecraft. The landing mission phase was especially constraining due to the high delta-Vs. After investigation, two propulsion systems were chosen by the team: nuclear ion-electric for the orbiter and solid rocket motors for the lander.

5.1 Orbiter Segment

The Orbiter uses nuclear ion-electric propulsion for a higher specific impulse between 4000-7000 sec compared to chemical propulsion, solar sails, tether propulsion, laser propulsion, and others. This choice allows reduction in the total mass of the propulsion system since less propellant is needed. Nuclear ion-electric propulsion is nearly 5-10 times more efficient than chemical propulsion systems [10]. Also, the nuclear ion-electric propulsion system has higher efficiency, making it significantly smaller. Nuclear ion-electric systems work on the concept of transforming energy from fission to electrical energy to produce thrust by accelerating the ionised propellant. For the Orbiter, a specific impulse of 4000 sec is estimated to be sufficient to perform the mission. Table 4 shows the propulsion summary for the Orbiter.

The downsides of nuclear electric-ion propulsion are the lower thrust and longer durations required to manoeuvre, resulting in an overall longer mission duration, with the journey to and from Mercury lasting around 5 to 7 years each way. However, since most of the cruise phase does not require high delta-V manoeuvres and a long mission duration is acceptable, nuclear ion-electric was determined to be the best option.

Table 4. Propulsion Summary for Orbiter

	Delta V(km/s)	Propellant Burned (kg)
Phase 1: Cruise	2.9	258
Phase 2: Landing Operations	-	-

Phase 3: Return Cruise	2.9	59
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The propellant burned for the return phase is reduced for two reasons:

1. Around 258 kg of propellant was burned during the trip to Mercury.
2. The Lander mass and associated solid rocket motors, in addition to the Ascent Segment's solid motor, are no longer present for the return trip.

The delta-Vs are from the JHU Decadal Survey [2] and were used to calculate the propellant burned using the Tsiolkovsky Rocket Equation:

$$\Delta v = v_e \ln \frac{m_0}{m_f} = I_{sp} g_0 \ln \frac{m_0}{m_f} \quad (1)$$

5.2 Lander and Ascent Segment

Landing on Mercury is a big challenge because of the estimated high delta-V requirement of about 3.1 km/s. Therefore, several high thrust, high mass fraction propulsion systems were considered, including nuclear propulsion, chemical propulsion, and solid rocket motors.

While nuclear and chemical propulsion provide powerful capabilities, their masses and sizes are too high for the Lander. Furthermore, both of these options also require tanks that add more mass and require more volume. They are more suitable for larger, longer duration missions. While tank material technology is improving exponentially, the sheer volume of the required propellant would cause the Lander to be considerably larger. For these reasons, solid rocket motors were considered favourable. Additionally, solid rocket motors can be discarded after use, allowing mass to be shed during the return phase making it more feasible. Tables 5 and 6 below show a propulsion summary.

Table 5. Propulsion Summary for Lander

	Delta V(km/s)	Propellant Burned (kg)
Phase 1: Cruise	-	-
Phase 2: Landing Operations (Descent)	3.1	1400
Phase 3: Return Cruise	-	-

Table 6. Propulsion Summary for Ascent Segment

	Delta V(km/s)	Propellant Burned (kg)
Phase 1: Cruise	-	-
Phase 2: Landing Operations (Ascent)	3.1	290
Phase 3: Return Cruise	-	-

The Lander segment uses four solid rocket motors with a propellant mass of about 350 kg. This propulsion capability allows the Lander to land efficiently on the surface of Mercury. Additionally, airbags could be employed to ensure a soft landing. However, a concrete decision was not made since an additional investigation was required to determine if this would be necessary.

In contrast, the Ascent Segment uses one solid rocket motor with a propellant mass of about 290 kg. This propulsion capability allows the Ascent Segment to detach from the Lander after the sample has been collected and jettison to rendezvous with the Orbiter in the parking orbit. However, since the Ascent Segment can detach from the Lander, it will require a high mass fraction to allow it to overcome the delta-Vs required for ascent.

5.3 Propulsion Subsystem Functional Overview

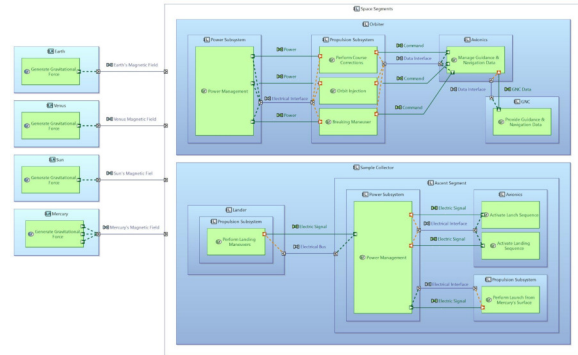


Fig. 10: Propulsion System Functional Overview

As previously discussed, all three mission segments have a propulsion subsystem. Nevertheless, their respective functions are different. For example, the Orbiter requires its propulsion system to perform correction and impulsive manoeuvres to increase or decrease the spacecraft's velocity to leave and enter a specific trajectory, all essential functions during the cruise phase. On the other hand, the Lander and the Ascent Segment are more robust since solid motors cannot be turned off once activated. This aspect also changes their respective configurations since, in the

Orbiter, the avionics subsystem can directly send commands to the propulsion subsystem based on the Guidance, Navigation and Control (GNC) Data. However, for the Lander and Ascent Segment, the avionics use the power subsystem to activate the motors using electrical signals. Lastly, as discussed before, the main external factors that affect this subsystem are the different gravitational environments that all segments encounter during the mission.

6. Guidance Navigation and Control

GNC refers to trajectory determination, sensing and controlling the spacecraft. Guidance is determining the necessary trajectory from the spacecraft's current position to the desired target by calculating the adjustments in velocity, rotation and acceleration needed to achieve a correct course. Furthermore, navigation is defined as the determination of the spacecraft's velocity, location and attitude at any given moment. Lastly, control is the manipulation of the spacecraft's steering controls, thrusters, etc., required to achieve guidance commands while maintaining vehicle pointing and stability. This subsystem is thus responsible for the spacecraft's orientation, executes Trajectory Correction Manoeuvres and analyses sensor data for hazard avoidance. It also regulates the orientation of the spacecraft [11]. For each subsystem, there is a corresponding GNC equipment as well as control modes for the particular segment it will have to execute.

6.1 Orbiter Segment

For the Orbiter, GNC's main objectives are to control the direction in which the Orbiter is pointing while keeping track of the relative position of the Orbiter to celestial bodies such as stars and the Sun. This information allows the Orbiter to correctly perform the needed manoeuvres while maintaining antennas pointed towards the Earth to maintain communication [12].

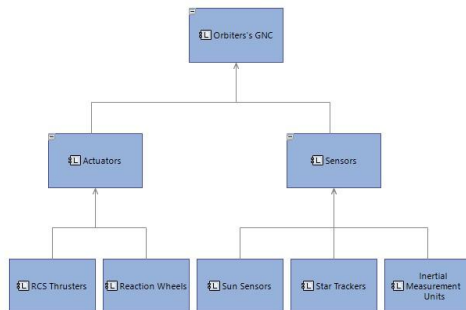


Fig. 11: Orbiter's GNC Logical Decomposition

6.1.1 Sensors

Estimating attitude combines the readings of three sensors: Sun sensors, star trackers and Inertial Measurement Units (IMU).

A 4-Quadrant DSS (Digital Sun Sensor) on the orbital stage is deployed to estimate the Satellite-to-Sun vector (Field of View (FOV): 140°, with up to 0.1° of accuracy) [13].

Two orthogonally mounted Sagitta Star Trackers (FOV: 25.4°, updating at a frequency of 5Hz) provide accurate information about the orientation of the Orbiter, using the stars as a reference and providing important attitude and attitude rate knowledge. The second star tracker is a backup in case the first one fails [14].

IMUs are used for attitude propagation between star tracker updates and when the attitude rate is higher than the star tracker limit. This design utilises an IMU equipped with four gyroscopes and four accelerometers to be deployed (with a bias stability less than 0.05 deg/h, 1000 µg, respectively) [13].

6.1.2 Actuators

The actuation in orbit relies mainly on Reaction Control System (RCS) thrusters and reaction wheels.

Four reaction wheels (torque: 100 Nm) are used in attitude control during orbiting. Four are used instead of three to provide fault tolerance. Fault tolerant control is the ability to tolerate faults and keep the control performance in the ideal range in the presence of faults [15][18].

4 RCS steering thrusters are to be used (375 N each), managing momentum and TCMs (trajectory correction manoeuvres) are delivered using short pulses (5 Hz) [13].

6.2 Lander and Ascent Segments

Mercury's extremely thin atmosphere, low gravity (twice the Moon's) and rugged terrain require a dynamic and precise GNC system to land safely. The GNC system will determine a landing location with the most level terrain and with the least amount of topographic hazards. It will additionally determine the states (velocity, attitude angle, and attitude-angle rate) that will result in a safe landing. Thus, the Lander should be able to recognize and avoid potential surface hazards since current Mercury surface data collected from previous missions is not very high fidelity as is required for landing.

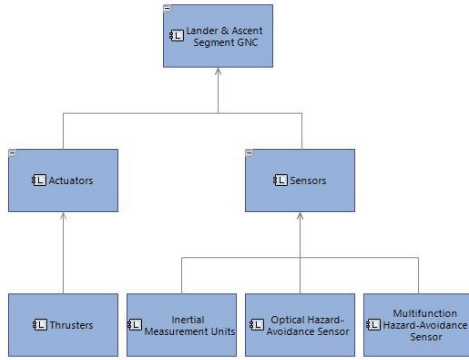


Fig. 12: Lander & Ascent Segment's GNC Logical Decomposition

6.2.1 Sensors

Two IMUs each consisting of three degrees of freedom (3-DOF) accelerometers and 3-DOF gyroscopes are deployed to calculate, respectively, the specific acceleration and angular velocity. During the landing phase, the IMUs are used for inertial navigation. They additionally signal key event triggers.

The accelerometers' measuring range is $\pm 20g$ where $g=9.8 \text{ m/s}^2$. The gyroscopes' measuring range is ± 450 ($^\circ$)/s [16].

6.2.2 Landing Radar and Hazard-Avoidance

The radar that will be used is the Terminal Descent Sensor (TDS), a Ka-band (35 GHz) Doppler radar with 6 narrow beam antennas. Each beam can simultaneously measure the slant range and ground-relative velocity along its direction [13][17].

Hazard avoidance comes in two steps with two different sensors. The first sensor, called Optical Hazard-Avoidance Sensor (O-HAS), consists of an optical imaging lens (FOV: 30°) that selects a rough landing spot using 2D images of the area during deceleration.

For precise hazard avoidance, the second sensor called the Multifunction Hazard-Avoidance Sensor (M-HAS) captures 3D data of the previously designated area and selects a safe landing spot during hovering. M-HAS consists of an identical optical imaging lens and a laser imaging lens (FOV: 30° for both) [16].

6.2.3 Actuators

For attitude control, 4 thrusters provide 250 N per thruster. The thruster pulse is 5 Hz, offering great precision [16].

6.3 GNC Computer

The GNC computer gathers and analyses information from sensors, actuators, and onboard data-handling systems, conducts GNC calculations in real-time, and delivers commands respective to the

current mission phase and helps identify and thus avoid potential hazards.

6.4 GNC Functional Overview

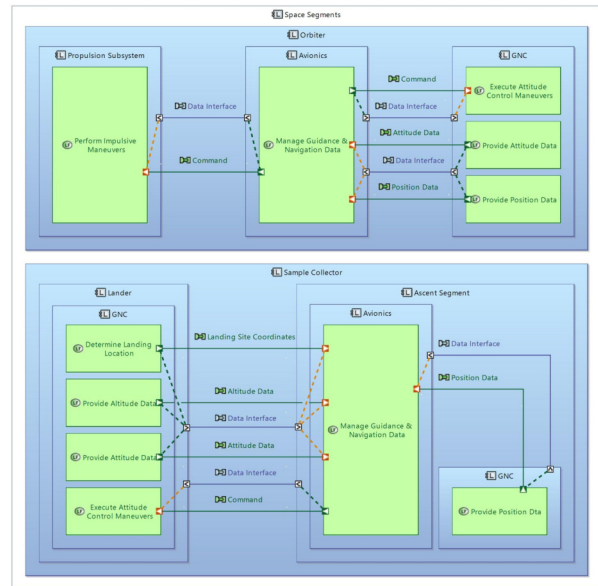


Fig. 12: GNC Subsystem Functional Overview

As previously discussed, the main functions of the GNC subsystem for all segments include altitude and attitude determination, and execution of correction manoeuvres to keep each segment stable on all axes. For the Orbiter, the propulsion subsystem performs the manoeuvres that affect the trajectory based on data provided by the GNC sensors. At the same time, the GNC actuators execute the manoeuvres for attitude control. For the Lander, the GNC subsystem determines the landing side and maintains the attitude of the Lander stable during landing. Finally, after the Ascent Segment separates, the GNC subsystem provides position data so that the Orbiter can capture it.

7. Electrical Power Subsystem

The power system supplies, regulates, distributes power and stores energy for the spacecraft subsystems and instruments through all the mission phases. There are 5 main components of the power system; the Next-Generation Radioisotope Thermoelectric Generator (Next-Gen RTG) as the spacecraft main source of power, the alkali fuel cell battery for the Lander-Ascent Segment power supply, the shunt regulator unit (SRU), the external power dissipation shunts and the propulsion diode box (PDB). A summary of the power system specification is given in Table 7 below.

7.1 Power Generation

Mercury's extreme hot and cold temperature coupled with the radiation threat of solar energetic particles (SEPs, which have a flux density about 10 times with respect to Earth's location), requires a compact, rugged power source that can provide power reliably under these harsh conditions. RTGs can operate efficiently under high temperature without degradation (unlike solar arrays) and are resistant to high energetic particle influence.

The power generation system contains two main elements: the SMRTG assembly and the converter. SMRTG—Segmented-Modular RTG is a modularized generator with segmented thermoelectric couples that can only operate in vacuum—was selected for the primary power generation for this mission and is the most powerful of the SMRTGs studied. It is composed of 16 modules that secure the plutonium in case of a launch accident. The thermoelectric couple (TEC) configuration 1 (TC-1) [19] was used to convert heat generated by radioactive decay of the plutonium heat source in the SMRTG into electricity.

At an efficiency of 16.5%, the power output of the SMRTG-16 (TC-1) is 590 W at the beginning of life (BOL), 558 W at beginning of mission (BOM), with a mass of 62 kg and an operating voltage of 32 V. The remaining thermal power output is radiated into space during flight and a cooling system maintains the temperature of the RTGs. The SMRTG-16 has a system power degradation average of 1.6% per year.

The power system will use two SMRTG-16s, both yielding power outputs with the cold-end temperature maintained at ~550 K, which reduces the efficiency to about 14.76% while the hot-end is still maintained at 1275 K. The operation details of the combine SMRTGs are shown in Table 7 below, where BOL is the beginning-of-life, BOM is beginning of mission, EOM is end of mission, W_{th} is thermal inventory at BOL, diameter w/fins is the diameter of RTG with fins [20].

Table 7. SMRTG-16 (TC-1) Specifications

P_o (BOL)	1180 W_e
P_o / Q_o (BOL): Efficiency	14.76%
P_o / m_{rtg} : Specific Power	9.47 W_e/kg
P (BOM)	1123 W_e
P (EOM)	935 W_e
Q_o	4000 W_{th}
Length	1.0372 m
Diameter w/fins	0.6960 m

Table 8 describes the power of the SMRTGs at the beginning of mission power, its power at Mercury and the regulated bus voltage. Further breakdown of the power summary at orbiter and landing stage is given in Table 9 and 10 respectively. In Table 10, the structure power includes power needed for the instruments aboard. Table 9 is for the Orbiter during cruise, parking

orbit, and return mission phases. Table 10 is for surface operations for the Lander and Ascent Segment. The Lander and Ascent Segment are assumed to be powered down during cruise and return. During return, the Ascent Segment receives power from the Orbiter to transfer the collected sample to the Deep Space Gateway.

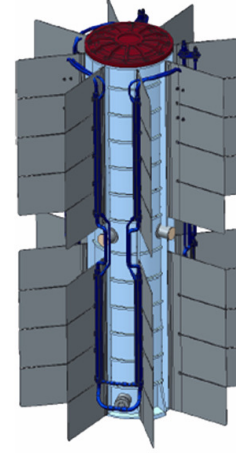


Fig. 13: Next Generation RTG concept.

Table 8. Spacecraft Power System Summary

RTG beginning of mission power	1123 W
RTG power at Mercury	1029 W
Regulated bus voltage	32 V

Table 9. Power Summary for Subsystems for Orbiter

Structure	5 W
Thermal	300 W
Power	40 W
Propulsion	500 W
CDH	21 W
GNC	70 W
Communications	67.5 W

Table 10. Power Summary for the Lander and Ascent Segment

Structure	310 W
Thermal	100 W
Power	20 W
CDH	21 W
GNC	15 W
Communications	1 W

7.1.2 Power Regulation

The Orbiter, Lander and Ascent Segment need external resistive shunts to dissipate excess power from the SMRTGs into space. During launch, there is little or no electrical power needed, so the shunt load would be the entire SMRTG early during the cruise phase. It can also be dissipated into the internal spacecraft heaters, if needed. The SRU contains three loop controllers to

control the 32 sequential analog shunts which keep the load constant on the SMRTG during spacecraft operational mode changes. One full shunt can dissipate 19.5 W. The shunts are designed to be $n + 1$ redundant with a full SMRTG fuel and can be disabled individually if there is a fault. The spacecraft bus voltage is set at 32.25 V to be slightly above the maximum power transfer level of the SMRTG because of voltage drops during distribution of power to individual loads. In case of short duration current surges during load turn-on or fault condition, there is a capacitor bank in the SRU.

7.1.3 Power Distribution Unit and Propulsion

The PDU is made up of a fully redundant solid-state power switching, pulsing, monitoring and fault protection features for the spacecraft. The block redundant power switching units provide low-voltage power distribution similar in concept and architecture to previous missions. Only the thruster electronics require power switching for the high voltage bus and are included as part of the high voltage shunt regulator.

7.2 Lander Power and Energy Storage

The presence of fuel cells in space programs started with the Gemini Space Program which focused on studying human presence in space. Today, decades after the Gemini missions, fuel cells have been improved continuously and are starting to be used in different industries. For this mission, alkaline fuel cells are used to provide energy storage for the landing, surface operations, and ascent. The fuel cells are located onboard the Ascent Segment but are also used to power the Lander as needed. The fuel cell is the most energy efficient device, converting chemical energy to electricity [21]. The fuel cell will provide clean power of 3.5 kW at an efficiency of about 60% and weighs about 50 kg.

The electrolyte in the alkaline fuel cell is aqueous potassium hydroxide soaked in a porous matrix. The anode catalysts can be Pt, Pd or Raney Ni with the cathode catalysts being made of Pt, Pd, Ag or MnO_2 . It operates in temperatures $<100^\circ\text{C}$ and has a lifetime of >5000 hours, making it perfectly usable for the Lander and Ascent Segment [22]. Table 10 shows the power allocation to the subsystems that are in the lander.

For the Lander, the fuel cells will aid with the collection of the samples by powering the PlanetVac system as previously described.

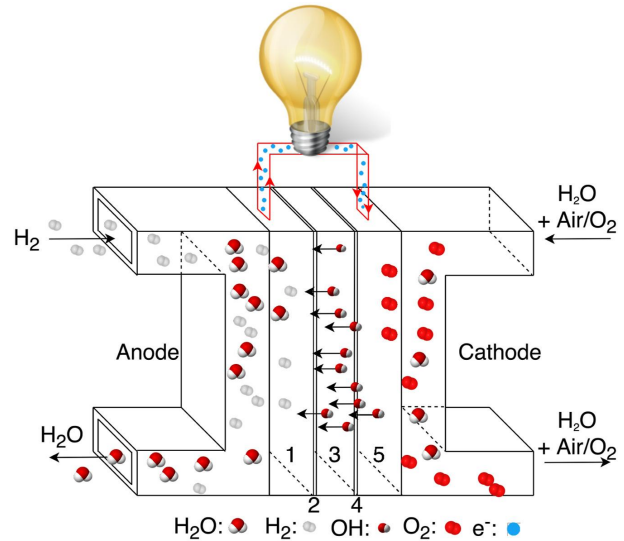


Fig. 14: Alkaline Fuel Cell

7.3 EPS Functional Overview

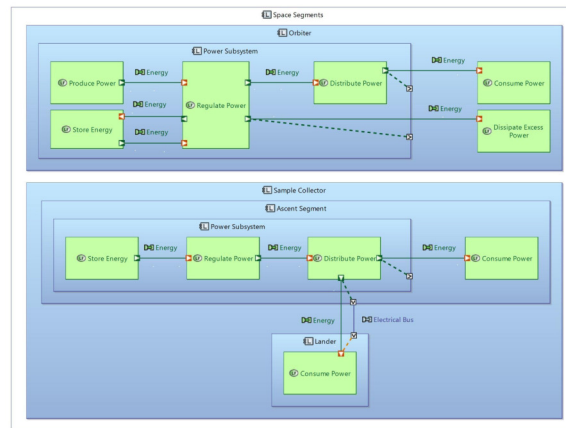


Fig. 15: EPS Subsystem Functional Overview

The EPS subsystem interacts heavily with the other subsystems. As previously discussed, the main functions of the EPS include power production, regulation distribution, and energy storage. However, in the case of the Lander and Ascent Segment, since they use a fuel cell as the primary power source, they both rely on the energy stored to perform their surface operations. On the other hand, the Orbiter has a continuous source of power, but it also has an added function of dissipation of excess power through SRUs as a result of having an RTG.

8. Thermal Control

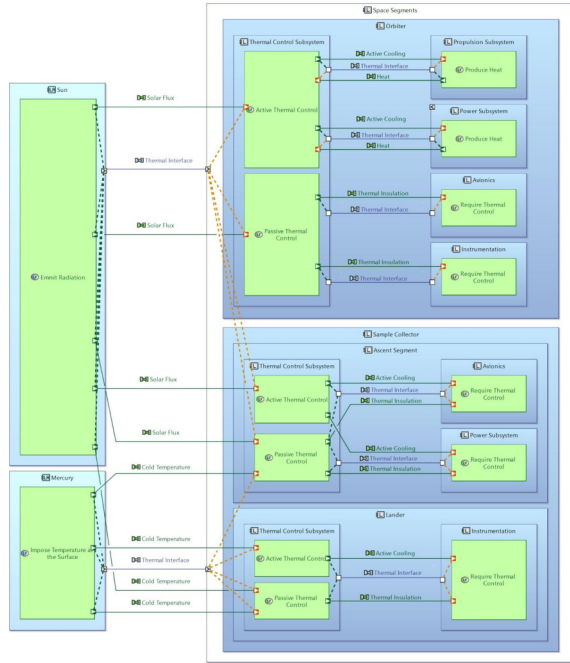


Fig. 16: Thermal Control Functional Overview

The Thermal Control Subsystem is responsible for ensuring that the ranges of temperature that each component experiences throughout the mission remains within acceptable operating ranges. Doing so requires awareness of the expected sources of heat from the external environment and those from inside the spacecraft.

Thermal control can be done actively or passively (without consuming power) depending on the thermal constraints. In this case, a mission to Mercury exposes all segments to constantly increasing solar flux, a significant source of radiative heat transfer. At the same time, the propulsion system and the power subsystem are significant heat sources within the spacecraft that require active cooling. In addition, other components, such as the avionics and instruments, need insulation because they are sensitive to high temperatures. In the case of the Lander and Ascent Segment, the components in those segments are more likely to experience low temperatures because of the landing location. The estimated ranges require passive and active methods of thermal control to insulate and sufficiently heat the avionics and instruments to ensure both segments can operate during surface operations.

8.1 Thermal Environment

As the closest planet to the Sun, Mercury's thermal environment is extremely hot. Mercury has a thin atmosphere with a pressure of approximately 10-15 Pa. Due to this extremely low atmospheric pressure, it is generally accepted that Mercury has no atmosphere and any atmospheric effect is ignored. Mercury's orbital

period is around 88 Earth days while its rotational period is around 58 days. A "Mercury day" lasts around 176 Earth days, resulting in a "Mercury night" of 88 Earth days. Due to the largely varying solar fluxes, Mercury has a maximum surface temperature of 427 °C and a minimum surface temperature of -173 °C, with an average surface temperature of 179 °C [23]. Due to Mercury's tremendous solar fluxes and high surface temperatures, the planet's Infrared emission ranges between 6 and 12,700 W/m². Although Mercury's average albedo is 0.12, its value ranges from 0.08 to 0.25 depending on the planet's geological features.

Table 11. Mercury Orbital Environments

	Perihelion	Aphelion	Mean
Direct solar (W/m ²)	14,462	6,278	9,126
Albedo	0.12	0.12	0.12
Planetary IR (W/m ²)			
Max (subsolar peak)	12,700	5,500	8000
Min (dark side)	6	6	6

The Orbiter is set to be placed in a 200 km x 200 km circular orbit to minimise the delta-V of landing. A "reference sphere" was introduced to indicate the temperature environment at various orbital positions of this orbit. The reference sphere is an isothermal perfect blackbody with a coefficient of emission/absorption of 1 [24]. The equilibrium temperatures of the reference sphere at the relevant orbital points are listed in Table 12.

Table 12. Reference Sphere in a 200 km x 200 km Orbit around Mercury

	Perihelion	Aphelion	Mean
$\beta = 0^\circ$			
Max (°C)	315	204	251
Min (°C)	-204	-204	-204
$\beta = 90^\circ$			
Max (°C)	244	147	187
Min (°C)	244	147	187

8.2 Thermal Materials/Decisions Chosen

Based on the temperatures in Table 12, it is evident that a $\beta = 90^\circ$ orbit for the orbiter provides the minimal range of temperature variation during orbital positions. Further, it eliminates the need for additional heaters. Because the mission is only for a limited time, it is also best to reach Mercury during its aphelion, when it is furthest from the sun.

8.3.1 Orbiter Thermal Control Design

The MESSENGER and BepiColombo missions inspired the thermal control design of the Orbiter. To protect the Lander and Ascent Segments during the cruise phase, a sun shield based on the MMO Sun shield and InterFace Structure (MOSIF) of the BepiColombo spacecraft was utilised [25]. The sun shield will be discarded after the Ascent Segment re-docking is complete at the end of the ascent phase. Inspired by the MESSENGER mission, a second sun shade was developed to shield the Orbiter from solar light. [26]

A radiator with suitable mass allocation for heat pipes was introduced, inspired by the Mercury Planetary Orbiter (MPO) segment of BepiColombo. The most significant technological development derived from the MPO radiator was the use of curved fins to reflect Mercury IR radiation while emitting thermal energy from the spacecraft to the environment (see Fig. 17) [27].

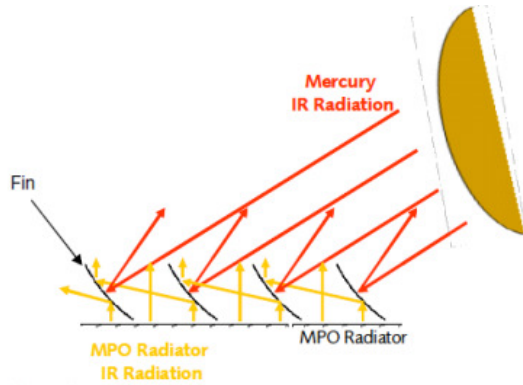


Fig. 17: MPO Radiator Fin Design [27]

The two SMRTG modules will be installed at the bottom of the Orbiter at a 120° angle on two sides of the hexagonal structure. The SMRTGs are thermally isolated from the main structure and a few thermal switches were added to each as a heater alternative. Each SMRTG is equipped with two shields. One is installed above the SMRTG to shield the spacecraft from its radiation. The other shield is rotating at an axis that matches the SMRTG's cylindrical axis. This enables the maintenance of the SMRTGs at an ideal temperature by regulating the Planetary IR radiation received.

As the high-temperature multi-layer insulation (HT-MLI) was flight-proven in the BepiColombo Mission, it was chosen to cover the Orbiter surfaces minus the radiator [27].

8.3.2 Lander Thermal Control Design

Thermal analysis of Mercury Lander concepts derived from decadal surveys demonstrates the possibility of landing on Mercury's surface during dusk. (up to three days before sunset) [25][26]. Sun incident angles and surface temperatures were found to be optimal for the operation of the lander, suggesting that

relatively minimum thermal management would be required for the two-day mission if the landing takes place during dusk. Since Mercury's surface geological features are only known to a resolution of up to a few metres, both the Ascent Segment and the Lander were fitted with thermostatically controlled heaters to ensure their survival in the event of absence of light from the Sun. The design of the Lander and Ascent Segment minimised the structure and payloads chosen; only the PlanetVac systems and a magnetometer were chosen. Additionally, most electronics, batteries, and other components are contained within a small compartment in the Ascent Segment. This co-location reduces the heater load compared to a more spread-out approach. It is expected that these design decisions will lower the amount of thermal energy required from heaters. If adequate solar energy is received during landed operations, the mission duration could be additionally prolonged.

The Ascent Segment and Lander are covered with MLI to protect the systems from high energy radiation during the entirety of the mission. Since the Lander and Ascent Segments are shielded from the intense solar energy by an Orbiter-mounted sun shield, standard MLI was selected for the system.

9. Avionics

The avionics subsystem is the backbone of the spacecraft and carries out all control operations providing critical command and data handling (CDH), electrical power system (EPS), and the attitude determination and control systems (ADCS) required to be able to reach or accomplish the mission objectives. As a result, the avionics system's reliability is crucial.

9.1 Orbiter Command and Data Handling

The CDH subsystem of the Orbiter will perform onboard operations and internal communications. The CDH subsystem requirements have been divided in two parts, one specifically for the commands and the other for the data handling.

9.1.1 Command Requirements

The Orbiter shall be capable of performing all of the required functions through the use of the on-board stored commands and the Orbiter bus shall accept and process real-time commands (RTC) and store-sequence commands (SSC) from the ground.

Table 13. Command Requirements

Real-time commands (RTCs)	Store-sequence command (SSC) files.
1. An immediate action should be taken by the spacecraft bus when receiving RTCs.	

2.	The Orbiter should be capable of autonomously executing a time-ordered series of commands or the SSC file.
3.	RTCs may command software or hardware activities as well as activities through the same subsystem.
4.	Only when the Orbiter validates receipt then the system should proceed to execute the command
5.	The orbiter will have stored on-board, in a library a script with the time-ordered commands which could be used by autonomous spacecraft housekeeping and fault protection functions.

9.1.2 Data Handling

The Orbiter data-handling system will be performing all data-handling functions for the spacecraft including:

1. Telecommand distribution throughout the spacecraft
2. Telemetry data collection from the spacecraft and data storage
3. Housekeeping or health status of the spacecraft
4. Functions with timing which includes distribution of time and keeping information synchronised

The data handling system is based on a dual-processor architecture with communication links, including ESA's own version of the Milbus ECSS-E-50-13C which is also used by BepiColombo, and high-rate IEEE 1355 serial data links [31][33]. The ECSS-E-50-13C bus is the data highway for data acquisition for all the platform units and subsystems with a high data rate and for command distribution via the Remote Terminal Unit (RTU).

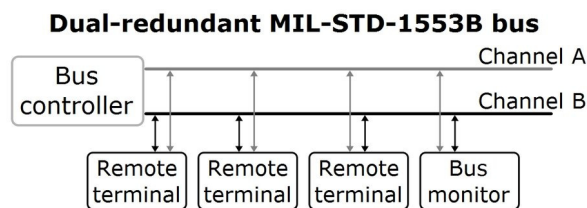


Fig. 18: Mil1553 Network Architecture

The IEEE 1355 links are used between both the Command and Data Handling Management unit (CDMU) processor and the Solid-state Mass Memory (SSMM) as well as the CDMU processor and GNC. The data management system includes two identical Processor Modules (PM) located in two fully redundant CDMU units. One PM is allocated to the CDMU (in charge of managing the rest of the spacecraft subsystems) and the other is allocated to the Ascent Segment (for power distribution and CDH).

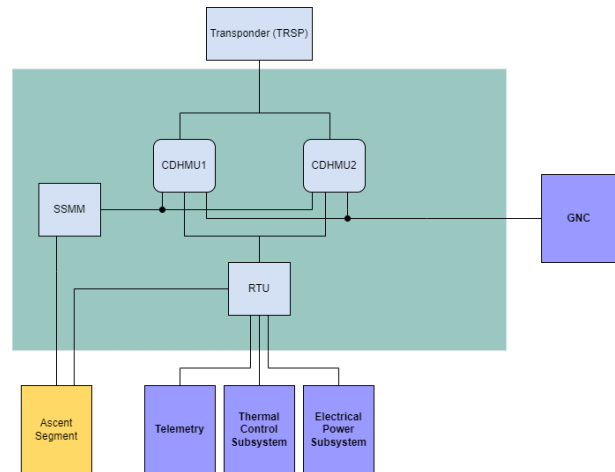


Fig. 19: Orbiter System Architecture

Telemetry is an important aspect of a space mission because it is what helps to keep track of the health of the spacecraft. For example, during the cruise and orbit phases, the Orbiter's data handling system will send data on the performance and health of the spacecraft [36].

Additionally, when designing the mission it was also important to consider potential failures, therefore, emergency commands were also added for fault protection. It is expected that, in the event of a failure that may interrupt or disrupt the normal ground-to-spacecraft communications, the Orbiter must provide autonomous on-board actions and run the stored command lines or SSCs [29][35].

9.2 Ascent Segment Avionics

The avionics system for the Ascent Segment is primarily responsible for two functions: CDH and power distribution. This consists of a Power Distribution Board, RCS controller, flight computer which is one of the fully redundant CDMUs, and internal batteries. All of the flight operations are conducted autonomously. The flight computer is the brain of the vehicle and is responsible for executing all algorithms and commanding the vehicle [34].

Due to mass and power restrictions, and the tremendous distance between Earth and Mercury, it is expected that the ground operators will not be able to receive telemetry signals or data on the health of the Ascent Segment. Yet, the Orbiter will know when the sample has been collected through an LED Array. After the sample collection is complete, an LED Array will light up allowing the Orbiter to visually recognise that sample collection is complete using cameras on-board. After this, the Orbiter will be able to position itself for the Ascent Segment. Since the two vehicles can't communicate with each other it is critical that the Orbiter moves to the proper location. After the sample

collection is complete and the LED Array is lit up, the Ascent Segment will wait for its timer of 2 days to complete before starting ascent phase operations. Once the Ascent Segment has docked, the Orbiter will send housekeeping data to the ground on both the Ascent Segment and Orbiter.

9.3 Lander Avionics

The Lander Segment fully relies on the Ascent Segment's avionics subsystems to keep with the modular approach.

9.4 Autonomous Operations

In general, for all three spacecraft segments the following autonomous operations shall be activated or provided for: a) normal autonomous operations, and b) autonomous fault protection.

For normal operations the spacecraft should provide for autonomous management of the following aspects:

1. Battery-State-of-Charge
2. Spacecraft Momentum
3. Thermal Control
4. Data Storage

For autonomous fault protection, in the event of a failure of the spacecraft health, the spacecraft should be capable of autonomously maintaining the minimum functions for safe system operations for at least 72 hours.

Specifically for the Orbiter, in addition to the above autonomous operations mentioned, during normal autonomous operations the orbiter should be capable of pointing the high gain antenna using a stored program.

During autonomous fault protection, the Orbiter telecommunications subsystem should initiate low-rate telemetry in the event of an on-board failure which may interfere with the high gain antenna communications.

10. Telecommunications

Even though there are three different spacecraft segments, for this mission design only the Orbiter will have a telecommunications subsystem. This decision was made to reduce the mass, power, and propellant needs of the Lander and Ascent Segment. The telecommunications subsystem includes one X-band medium gain antenna, one Ka-Band High Gain Antenna, a Small Deep-Space Transponder (SDST) (reliable X-band and Ka-Band), and a Ka-Band Solid State Power Amplifier [30][32].

Table 14. Orbiter Mission Operations & Ground Data System Summary

	Cruise/Return Phase	Parking Orbit Phase
Downlink		
Number of contacts per week	5	5
Downlink Frequency Band, GHz	8.4	8.4
Telemetry Data Rate(s), kbps	0.030-0.5	
Transmitting Antenna Type(s) and Gain(s), DBi	MGA, 10 - 16 dBi	HGA, 22 - 25 dBi
Transmitter Peak Power, Watts	20	
Downlink Receiving Antenna Gain, DBi	68.2	68.2
Transmitting Power Amplifier Output, Watts	11	
Uplink		
Number of Uplinks	5/week	3/day
Uplink Frequency Band, GHz	7.1 - 7.2	7.1 - 7.2
Telecommand Data Rate, kbps	0.030-0.1	0.1-0.4
Receiving Antenna Type(s) and Gain(s), DBi	MGA, 10 - 16 dBi	HGA, 22 - 25 dBi

11. Conclusions

Conducting a sample return mission to Mercury will truly change what is known about the universe. Additionally, the engineering advances that this paper alludes to will result in new technologies that will impact life on Earth.

By employing modularity, the challenges of power required and mass can be reduced. Having different components located in certain areas and discarding segments or components no longer needed is an example of this. Also by focusing on the primary mission goal of returning a sample, certain design decisions were made including instrumentation choices that further minimised the design. The propulsion system chosen for this mission is also novel and demonstrates a potential solution to the difficulties associated with deep space missions.

This paper has presented a high level view of what such a mission would look like and aims to overcome the difficulties with conducting a mission to Mercury.

By completing this mission, the scientific community will expand their knowledge of our universe and the formation of our own planet.

Acknowledgements

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Appendix A

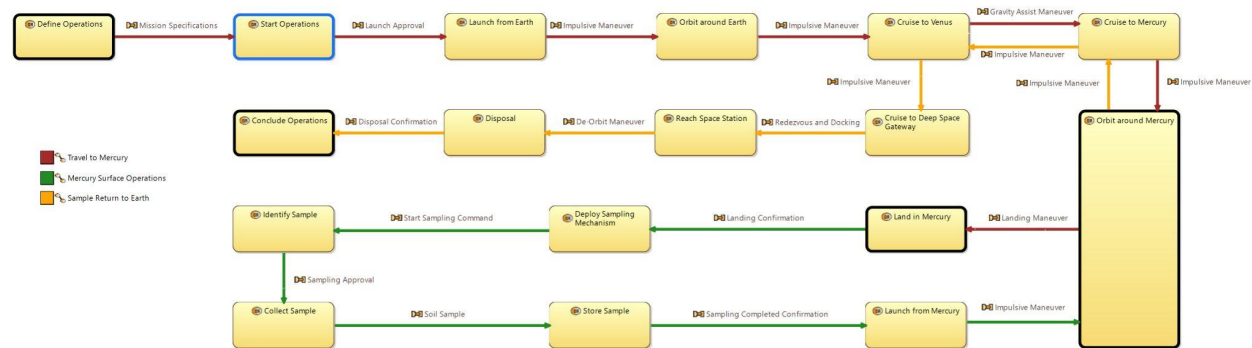


Fig. 20: Operational Activities & Interactions for Mission Execution

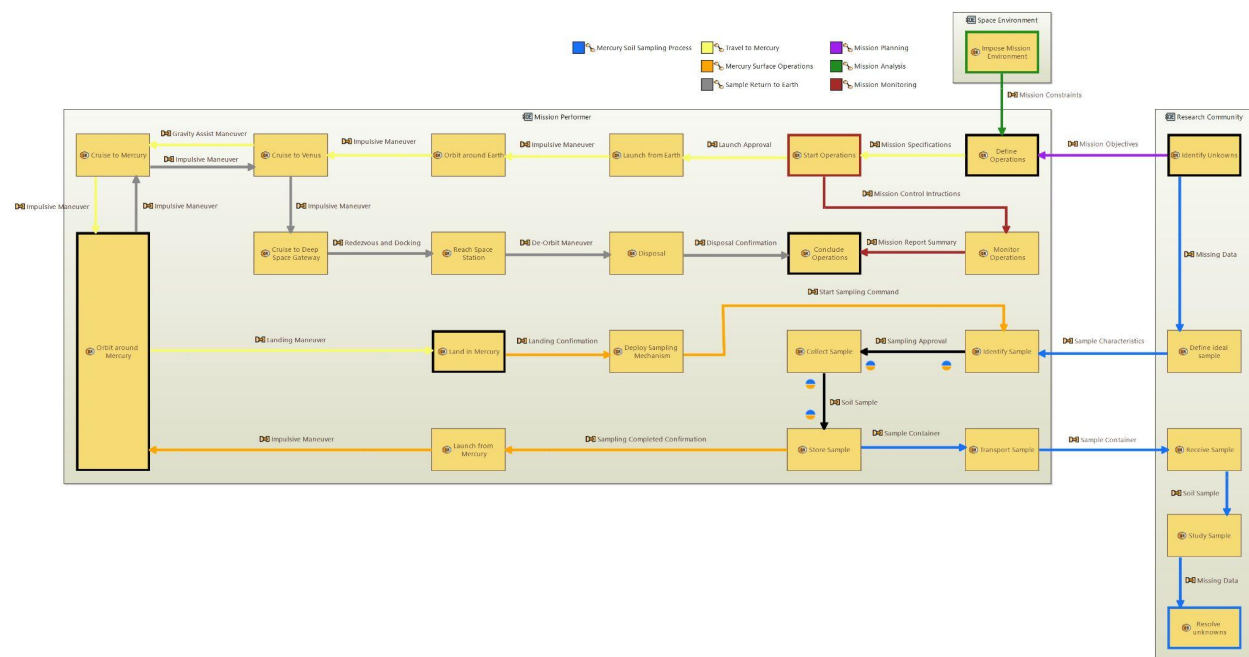


Fig. 21: Operational Architecture Displaying Operational Entities and Respective Activities and Interactions

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